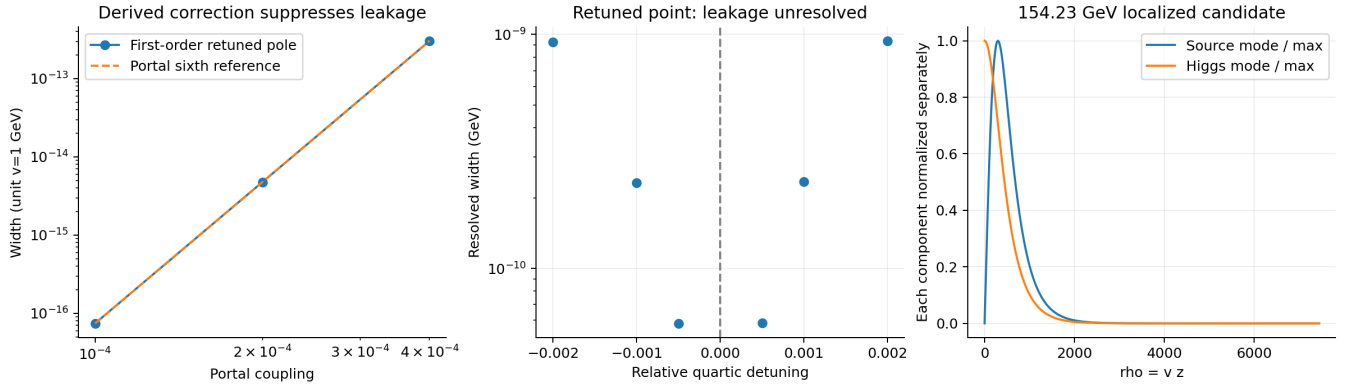


Retuned polynomial radiation nodes and localized wall-mode candidates

The wall resonance calculation now has a constructive continuation. Distortion of the Higgs background shifts the scalar-reference radiation node. An independently derived first-order threshold correction cancels the next radiation amplitude and changes the observed small-portal width scaling from the fourth to the sixth power. A joint nonlinear wall/eigenmode calculation then finds a curve of finite-radius localized embedded-state candidates. Retuning the source quartic while keeping the 246 GeV Higgs vacuum, Higgs quartic 0.13 and all other supplied source/mediator scales fixed gives a candidate near 154.23 GeV. This is a different source potential from the earlier 11.62 TeV resonance benchmark. Its leakage is unresolved at the tuned point and reappears under quartic detuning. Infinite-domain existence and protection under quantum running remain open.



The question resolved by this continuation. The previous chapter derived the leading radiation form factor $I(p) = \pi(1-p^2)/[2 \cosh(\pi p/2)]$. In the zero-Gaussian-coupling scalar reference, its node is $m_H^2 = \lambda$, where all masses here are divided by v^2 , $m_H^2 = 2 \lambda_H h_0^2$ and $h_0 = v_H/v$. This node is inside the radial-Higgs continuum because the shape energy is $3 \lambda/2$. The full coupled wall did not remain nonradiating at that uncorrected condition: its small-portal width instead approached κ to the fourth. The present work asks how the condition itself must move once the wall background and fluctuation operator are treated consistently. It does not assume that the earlier leading node was a complete bound state.

A controlled perturbative expansion. Set $g=0$, $k = \sqrt{\lambda/2}$, $x = k \rho$ and $u_0 = \tanh(x)$. Write the radial background as $h = h_0 + \kappa h_1 + O(\kappa^2)$. At the unperturbed node the exact background equation gives $(-d_\rho^2 + 2\kappa^2)h_1 = h_0 \text{sech}(x)^2$. Thus $h_1 = (h_0/\kappa^2)H(x)$, where the dimensionless response satisfies $(-d_x^2 + 2)H = \text{sech}(x)^2$. The canonical source background and its shape energy first change at order κ^2 . The shape mode normalized to source derivative one at the center is $\psi_0 = \text{sech}(x)\tanh(x)/\kappa$. Its leading induced Higgs mode is $\kappa \chi_1$, with $\chi_1 = h_0 \text{sech}(x)/\kappa^3$. Direct differentiation verifies $(d_\rho^2 + \kappa^2)\chi_1 = 2h_0 u_0 \psi_0$. This gives an explicitly localized first-order Higgs response despite the open channel.

The second-order radiation source. Let $\lambda_H = \lambda_{H0} + \kappa L_1 + O(\kappa^2)$, with h_0 fixed. Expanding the complete canonical Hessian gives the second-order Higgs forcing $F_2 = 2u_0 h_1 \psi_0 + [6 \lambda_{H0} h_0 h_1 + (u_0^2 - 1) + 2h_0^2 L_1]\chi_1$. The terms include both the background-induced variation of the source/Higgs mixing and the first-order distortion of the Higgs diagonal operator. Omitting either term would change the retuning coefficient. The real even outgoing radiation amplitude at this order is proportional to the cosine overlap of F_2 at the open wave number k . Its vanishing determines L_1 without minimizing a numerical width.

The resulting universal coefficient. Define $F = \pi/\cosh(\pi/2)$, and J as the real-line integral of $\cos(x)$ times $[(8-2 \text{sech}(x)^2)\text{sech}(x)H(x) - \text{sech}(x)^3]$. Then the compensating unmixed Higgs-threshold shift is $2\lambda_H h_0^2 = \lambda_H + C \kappa + O(\kappa^2)$, with $C = -J/F$. Both the length and amplitude factors cancel from C : it is independent of λ and h_0 within this $g=0$ perturbative problem. The calculated value is approximately -1.36468352107739 . The equality defining C is the perturbative derivation; its displayed decimal comes from numerical integration, rather than an exact rational or algebraic identification.

Independent evaluation of the response. One implementation solves the even Green-response boundary problem on x in $[0, 24]$, with $H'(0) = 0$ and $H'(24) + \sqrt{2}H(24) = 0$, then integrates J . A separate implementation uses the

explicit free Green convolution: for $x \geq 0$, $H(x)$ is the integral over $y \geq 0$ of $[\exp(-\sqrt{2})\text{absolute}(x-y)) + \exp(-\sqrt{2}(x+y))]\text{sech}(y)$ squared/[$2\sqrt{2}$]. It shares neither a boundary solver nor its profile interpolation. The two values of C agree to about one part in ten to the fifteenth in this environment, and shortening the response domain to 18 retains ten decimal places. Quadrature estimates and solver residuals are recorded separately; these comparisons are not certified error enclosures.

Cancellation observed at the next portal order. In the unit- v model with $\lambda=0.1$, $h_0=1$, $g=0$, set $\lambda_H=(\lambda+C\kappa)/2$ using only the derived first-order coefficient. Portals 0.0001, 0.0002 and 0.0004 give independently calculated outgoing widths approximately 7.43745 times ten to the minus seventeen, 4.75304 times ten to the minus fifteen and 3.03189 times ten to the minus thirteen GeV. The successive ratios are approximately 63.91 and 63.79, approaching the sixth-power factor 64. At the uncorrected node the corresponding first width was about 4.50524 times ten to the minus twelve GeV. The improvement comes from the analytically specified amplitude cancellation; these first-order-retuned quartics were not selected by fitting those outgoing widths. Higher coefficients and a certified asymptotic remainder have not been derived.

Solving the complete retuned condition. The new boundary solver treats the static three-field background and a real three-field fluctuation as twelve first-order variables, with two unknown parameters: the real mode energy and one positive quartic. Background parity and right vacuum values give six conditions. Odd-source/even-singlet/even-Higgs mode parity and unit source derivative give four. At the outer radius, both the open eigenchannel amplitude and derivative are set to zero, while the two closed eigenchannels obey decaying Robin conditions. These give four more conditions, matching the fourteen unknown degrees of freedom. The quartic is represented by the exponential of a free parameter to preserve positivity. The boundary system can retune either λ_H at fixed source λ , or source λ at fixed Higgs quartic. Its initial guess uses the scalar-node value of the chosen quartic; it is not an unchanged continuation of an arbitrary input source potential.

Numerical localized candidates and finite-domain meaning. For the unit- v , $\lambda=0.1$, $h_0=1$ example at portal 0.001, the full joint solve selects λ_H approximately 0.04934530452095 and energy approximately 0.15006245567178. The energy lies strictly between the open Higgs threshold and the closed source threshold. Matching radii of fourteen, eighteen and twenty-two initial source decay lengths, with tighter solver tolerances, give consistent quartics and energies. The open-channel endpoint constraints are satisfied to floating-point residuals; the closed components become exponentially smaller with radius. A stored exterior norm estimate assumes the operator is exactly its constant vacuum value beyond that radius. It demonstrates localization in the matched constant-exterior problem, rather than bounding the tail error of the actual infinite-domain operator.

An outgoing solver that does not impose localization checks the result separately. It receives only the retuned action and solves its ordinary outgoing pole problem. At the retuned point it returns tiny signed imaginary parts that vary near zero with domain and tolerance; the relative current-identity error also grows as both current and width approach the numerical floor. These outputs are recorded as unresolved, not converted into a huge lifetime or a rigorous upper bound. Relative Higgs-quartic detunings of minus and plus 0.001 instead restore resolved widths around 2.32370 times ten to the minus ten and 2.34138 times ten to the minus ten GeV. Halving and doubling the detuning produce the local quadratic leakage behavior visible in the scientific packet. Such sensitivity identifies a tuned embedded-state candidate, rather than a demonstrated symmetry-protected mode.

Retuning with the declared weak Higgs quartic. The scalar-reference node cannot be reached by changing only λ_H in the old $\lambda=0.1$, $v=30000$ GeV, $v_H=246$ GeV action while keeping λ_H perturbatively small: its zero-portal value would be approximately 743.6. The alternative is to change the source quartic. The joint calculation keeps $v=30000$ GeV, $M=300000$ GeV, current coupling $g=3000$ GeV, $v_H=246$ GeV, $\lambda_H=0.13$ and portal 0.0000001. It selects source λ approximately 1.76188164948 times ten to the minus five and gives mode mass approximately 154.2250242 GeV. This positive, small source quartic changes the source spectrum. The complete vacuum eigenchannel thresholds remain on either side of the candidate energy, so the radial-Higgs channel is open and the source channel closed. The finite Gaussian mixing is included directly rather than approximated by the $g=0$ coefficient.

The fixed-Higgs case also survives the domain comparisons. Radii of fourteen, eighteen and twenty-two initial scalar decay lengths give the source quartic and mode mass consistently. An independent outgoing calculation at the selected point has leakage unresolved near zero. Source-quartic detunings of minus and plus 0.001 restore widths approximately 2.09854 times ten to the minus twelve and 2.06787 times ten to the minus twelve GeV. Neither the new small source

quartic nor its retuned relation has been derived from the flavor symmetry or RG evolution. The earlier benchmark transition temperature, bulk Higgs-pair decay width, annihilation estimate and gravitational-wave template cannot be carried over unchanged to this new potential. Only radial spectator fluctuations have been included; gauge, angular-Higgs and fermionic channels can change its lifetime problem.

Delivered execution and relation to concurrent work. The module is `python/perfectpower/wall_embedded_states.py` and the driver is `python/develop_wall_embedded_states.py`. Run both the scientific driver and focused tests under `PYTHONPATH=python OPENBLAS_NUM_THREADS=1`; the test command is `python -m unittest discover -s python/tests -p test_wall_embedded_states.py`. The receipt contains the independently evaluated coefficient, first-order-corrected outgoing widths, the full quartic candidate curve, domain/tolerance comparisons, detuning poles and the fixed-Higgs source-quartic candidates. Plot with `python scripts/plot_wall_embedded_states.py`. Eight new tests check the independent Green calculation, domain convergence, open/closed thresholds, localization conditions, perturbative slope, sixth-power scaling, independent outgoing responses and fixed Higgs inputs. The prior forty-six related wall tests are retained. This work uses the wall lane while the incoming scalar quantum frame-feedback, singular residue-atlas and repeated-factor arithmetic commits remain intact. It introduces no new Lean theorem.

Remaining mathematics and physical interpretation. A genuine infinite-domain existence result needs control of the background and localized-mode errors, an asymptotic radiation-amplitude zero and a nondegeneracy or continuation argument. The present finite-radius equations and perturbative cancellation identify the candidate and its expected continuation, but do not supply that theorem. A protected flavor or cosmological mechanism would also need a reason for the quartic relation to persist under the appropriate quantum corrections and for the actual flavor vacuum to realize this spectator structure. The existence and parameter dependence of embedded modes are established research subjects; useful mathematical context is <https://arxiv.org/abs/2205.09951>. The result here is the derived compensating coefficient, its tested change in leakage scaling, and the joint numerical retuning interface on the repository's explicit canonical model.